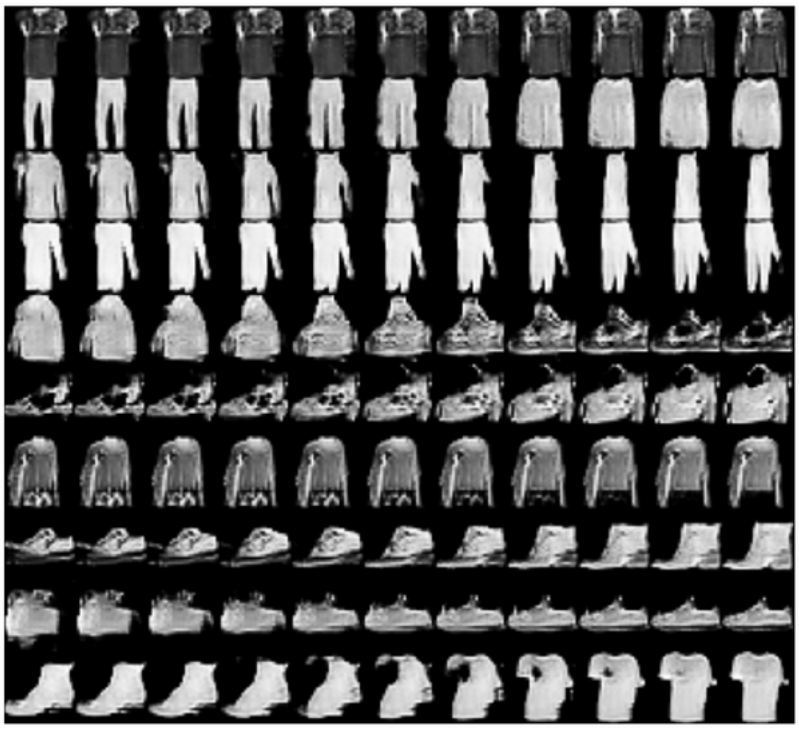
Lab 1 – MLP Autoencoder

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Part 3: Post-Lab



Step 1 – Bottleneck Size

1. Train two additional models, with bottleneck sizes of 8 and 128, using 30 epochs each.
2. Compare reconstructions of the same test images across all three models.
3. Plot example inputs and outputs of each model.
4. Record the final training loss for each model.

Step 2 – Bottleneck Interpolation

Create a new module, **`interpolate.py`**. Write code to follow these steps:

1. Select two images. Normalize and flatten as in **`test.py`**.
2. Pass each through the **`encode`** method from **`model.py`**, and return their two bottleneck tensors, **`z1`** and **`z2`**.
3. Linearly interpolate through these tensors for **`n`** steps. This means producing **`n`** new bottleneck tensors by taking weighted averages of the two:
4. **`z = (1 - a) * z1 + a * z2`**, where **`a`** takes **`n`** evenly spaced values from 0 to 1.
5. Pass all **`n`** tensors through **`decode`**, along with **`z1`** and **`z2`**. Reshape output to 28x28.
6. Plot all **`n + 2`** results in a single row, from two different input classes. Use **`n = 8`**.

Using **`MLP.32.pth`**, run **`interpolate.py`** and save output.

Step 3 – Report

The report should include a title, your name(s) and student number(s). If working in a team of two, only submit once in OnQ, but record both partners' information. Include the following sections in the report.

1. **Model:** This section should contain all details of the network, sufficient to recreate the model. This could be in textual form, diagram form, etc.
2. **Training:** This section should include all details of the training, sufficient to exactly reproduce the training. This should include the optimization method, optimization hyperparameters, any learning rate scheduling used, and all other relevant details.
3. **Outputs:** Include the outputs of Steps 1 and 2. This includes the bottleneck size comparison figures and the final training losses for each, along with the full row of bottleneck interpolation. Also include the outputs of **`test.py`** and **`denoise.py`**.
4. **Analysis:** This section should analyse the results of testing, denoising, bottleneck size, and bottleneck interpolation.

a. Within this MLP autoencoder, a 784-dimension image is compressed to 32 dimensions. Based on the reconstructions, which information survives and what is lost? Which classes reconstruct the worst, and why might this be?
b. Why does the MLP autoencoder structure work so well for noise removal?
c. How does bottleneck size affect the reconstruction quality?
d. Describe the images in the interpolation. Are they blends of two images, or something else? What does this suggest about the bottleneck?

Step 4 – Submission Format

Both the report, the complete working source directory and the self-assessment spreadsheet should be compressed into a single **<zzz>.zip** file, where filename **<zzz>** is replaced with your student number. If you are in a team of 2, then concatenate both student numbers, separated by an underscore.

All code should be executable within a terminal. Include a text file with the appropriate terminal lines. The zipped directory **should not** contain the FashionMNIST dataset. The zipped directory **must** include the following files:

- `model.py, train.py, test.py`
- `denoise.py, interpolate.py`
- `MLP.32.pth`
- `loss.MLP.32.png`
- Output of `interpolate.py`
- The post-lab report

The lab is worth a total of 8 marks. The marking rubric is as follows:

Item	Marks
In-Lab	3 (marked in lab)
Step 1 – Bottleneck Size	0.5
Step 2 – Bottleneck Interpolation	1
Post-Lab Report	2
Correct submission format	0.5
Total:	**7**